Instant Insanity with compact QAOA

- Four cubes. Four colours. Each tower side must show every colour once.
- CORE completed: full-puzzle circuit verified locally and on SV1.
- 12-qubit high-success circuit; original experiment used an extra marker.

Represent choices, not every cube rotation

- Each cube has three opposite-face pairs.
- Choose one pair for front/back and another for left/right: six choices.
- Three bits per cube encode those six choices; two bit patterns are forbidden.
- Classical cycle orientation turns the selected pairs into visible faces.

How QAOA searches the full puzzle

- Start each cube equally spread across its six legal choices.
- Cost: penalize colours appearing other than twice across each opposite-side axis.
- A cost layer changes phases; a reflection mixer lets choices interfere.
- Repeat three times. A classical optimizer tunes angles using mean cost.
- Known answers are used only for checks, never for circuit initialization.

Check the model before running it

- Exhaustively checked all 4,096 encoding strings against the classical cost.
- Exactly two strings have zero cost; both produce solved towers.
- Independent Qiskit and compiled statevectors agree.
- Aer executes actual measurements; all failures stay in the denominator.
- Every accepted candidate passes the original four-side checker.

Local success improvement and its cost

- Original Aer p=3: 131/4096 = 3.1982%.
- New Aer p=32: 4076/4096 = 99.5117%; exact 99.5380%.
- New circuit: 12 qubits; 18,757 two-qubit gates; depth 22,288.
- Direct success optimization and more layers raise ideal success; depth increases.
- The new circuit was also confirmed on SV1 in a separately approved job.

High-success Amazon Braket SV1 confirmation

- New SV1 p=32: 4073/4096 = 99.4385%; Aer: 99.5117%.
- 12 qubits; 18,757 two-qubit gates; depth 22,288.
- Actual charge: 0.375 credits. Two approved jobs total; no retries.
- New job ID: aws:aws:sim:sv1-4305-qjob-6ad3a9d138127b44c71b9f9d
- Original p=3 SV1: 3.0273%; both jobs and billing are in the README.

What we learned and what comes next

- Compact encoding halves the algorithm qubits compared with the starter.
- Preserving valid choices avoids spending shots on forbidden cube states.
- Saving qubits increases cost-layer gate complexity; noisy hardware is a challenge.
- Next: reduce cost-layer gates and compare mixers at fixed optimization budgets.
- AI assistance was disclosed and independently checked; no quantum advantage claimed.